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The effects of magnesium and vitamin E co-supplementation on wound healing and metabolic status in patients with diabetic foot ulcer: a randomized, double-blind, placebo-controlled trial

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ABSTRACT

This study was carried out to determine the effects of magnesium and vitamin E co-supplementation on wound healing and metabolic status in patients with diabetic foot ulcer (DFU). The current randomized, double-blind, placebo-controlled trial was conducted among 57 patients with grade 3 DFU. Participants were randomly divided into two groups to take either 250 mg magnesium oxide plus 400 IU vitamin E (n=29) or placebo per day (n=28) for 12 weeks. Compared with the placebo, taking magnesium plus vitamin E supplements reduced ulcer length [β (difference in the mean of outcomes measures between treatment groups) -0.56 cm; 95% CI, -0.92, -0.20; P=0.003], width (β -0.35 cm; 95% CI, -0.64, -0.05; P=0.02) and depth (β -0.18 cm; 95% CI, -0.33, -0.02; P=0.02). In addition, co-supplementation led to a significant reduction in fasting plasma glucose (β -13.41 mg/dL; 95% CI, -20.96, -5.86; P=0.001), insulin (β -1.45 µIU/mL; 95% CI, -2.37, -0.52; P=0.003), insulin resistance (β -0.60; 95% CI, -0.99, -0.20; P=0.003) and HbA1c (β -0.32 %; 95% CI, -0.48, -0.16; P<0.003), and a significant elevation in insulin sensitivity (β 0.007; 95% CI, 0.003, 0.01; P<0.001) compared with the placebo. Additionally, compared with the placebo, taking magnesium plus vitamin E supplements decreased triglycerides (β -10.08 mg/dL; 95% CI, -19.70, -0.46; P=0.04), LDL-cholesterol (β -5.88 mg/dL; 95% CI, -11.42, -0.34; P=0.03), high sensitivity C-reactive protein (hs-CRP) (β -

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3.42 mg/L; 95% CI, -4.44, -2.41; P<0.001) and malondialdehyde (MDA) (β -0.30 µmol/L; 95% CI, -0.45, -0.15; P<0.001), and increased HDL-cholesterol (β 2.62 mg/dL; 95% CI, 0.60, 4.63; P=0.01) and total antioxidant capacity (TAC) levels (β 53.61 mmol/L; 95% CI, 4.65, 102.57; P=0.03). Overall, magnesium and vitamin E co-supplementation for 12 weeks to patients with DFU had beneficial effects on ulcer size, glycemic control, triglycerides, LDL- and HDL-cholesterol, hs-CRP, TAC and MDA levels.

Keywords: Magnesium, vitamin E, wound healing, metabolic status, diabetic foot ulcer

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INTRODUCTION

The diabetic foot ulcer (DFU) is a chronic consequence of diabetes mellitus which mainly caused by neuropathy, different grades of ischemia, and infection that coincide with poor glycemic control and diminished neutrophil function.[1] The mortality rate among diabetic patients with DFU is higher than twofold compared with non-ulcer patients.[2] Based on the previous evidence, the effective management of diabetes and timely treatment of foot ulcer are implicated for decrease the morbidity and mortality of DFU.[3] The nutritional status of diabetic patients correlates with the severity of ulcer and play an important role in the reversing of wound risk.[4]

According to the prior studies, the alternation of magnesium metabolism is a common feature in diabetic patients which interference with glycemic control and contributes to impaired insulin sensitivity and chronic inflammatory status.[5] In addition, observational evidence shown that there is an inverse association between magnesium levels and the occurrence of foot ulcers, nephropathy and retinopathy.[6] Similarly, it is reported that diabetic patients with low antioxidant concentrations have a higher risk for experiencing latter complications. Micronutrient deficiencies, including vitamin E, vitamin C and magnesium have been implicated in several diseases associated with oxidative stress.[7, 8] In this context, circulating concentrations of vitamin E were noted to be significantly lower in subjects with DFU.[7] Previous studies have revealed that magnesium supplementation have beneficial effects on fasting glucose and some lipid profiles in both diabetic and non-diabetic individuals.[9] Our recent research showed that magnesium

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supplementation to DFU patients led to a significant improvement in ulcer size and metabolic status.[10] In a meta-analysis conducted by Xu et al.[11], vitamin E supplementation to type 2 diabetes mellitus (T2DM) patients with low vitamin E levels and poor glycemic control significantly improved HbA1c and insulin levels. Few evidence indicating an association between magnesium and vitamin E status in metabolic disease. For example, magnesium supplementation to patients with chronic fatigue syndrome led to a significant elevation in vitamin E concentrations.[12] In T2DM patients, vitamin E supplementation also increased intracellular magnesium levels.[13] In addition, it was reported that the co-administration of magnesium and vitamin E to obese and diabetic rat model improved lipid concentrations better than vitamin E [14] or magnesium alone.[15]

Magnesium is an essential cofactor for enzymes involved in carbohydrate and lipid metabolism and acts as anti-inflammatory agents via the regulatory influence on inflammation resolving pathways.[16] It is also necessary for the synthesis of protein and tissue development during wound curing process.[17] Experimental studies proposed that vitamin E has been effective in supporting diabetic wound healing by the regulation of inflammatory reaction, apoptosis and antioxidant defense system.[18] Considering the numerous beneficial effects of magnesium and vitamin E on mechanisms related to metabolic control, and wound healing process and due to the proposed additive action for the combination of these nutrients, the present study was conducted to evaluate the effects of magnesium and vitamin E co-supplementation on wound healing,

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glycemic status, lipid profiles, and biomarkers of inflammation and oxidative stress in patients with DFU.

METHODS

Trial design

The current investigation was a randomized, double-blind, placebo-controlled clinical trial.

Participants

The present study was conducted among 57 patients with grade 3 DFU according to "Wagner-Meggitt's" criteria [19], aged 40-85 years who were referred to the Shahid Beheshti Clinic in Kashan, Iran, between November 2017 and September 2018. Pregnant and breastfed patients, participants who consumed magnesium, vitamin E or other antioxidative supplements during past 3 months, taking anti-inflammatory agents, change in consuming medications throughout the study and patients with history of diseases which influence the development of DFU including chronic trauma, were excluded.

Ethics statements

All procedures were in agreement with the principles of the Helsinki Declaration; the study protocol was approved by the ethics committee of Kashan University of Medical Sciences

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(KAUMS) and has been registered at www.irct.ir (ID IRCT2017090533941N21). All participants provided informed consent formed to participate in the intervention.

Study design

Sixty participants (n=30 in each group) were randomized to take either 250 mg/day magnesium oxide (21st Century Pharmaceutical Company, Arizona, USA) plus 400 IU/day vitamin E (Zahravi Pharmaceutical Company, Tabriz, Iran) or placebo (Barij Essence Pharmaceuticals, Kashan, Iran) for 12 weeks. Due to lack of evidence about the appropriate dosage of magnesium and vitamin E for patients with DFU, we used the above-mentioned dose of magnesium based on a previous study in women with gestational diabetes mellitus [20] and dose of vitamin E based on a previous study in patients with T2DM.[21] Tolerable upper intake level for vitamin E and magnesium is 1000 mg and 350 mg per day, respectively.[22, 23] All participants underwent a similar treatment protocol for the Diabetic Foot, based on the Infectious Diseases Society of America (IDSA) (24). All patients completed 3-day food records at baseline, weeks 5, 9 and 12 of the intervention. Daily macro- and micro-nutrient intakes were analyzed by nutritionist IV software (First Databank, San Bruno, CA).

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Treatment Adherence

To determine the compliance, the remaining supplements were counted and subtracted from the amount of supplements provided to the participants. To increase compliance, magnesium levels were evaluated by enzymatic method.

Assessment of anthropometric measures

Weight and height of study participants were determined in an overnight fasting status using a standard scale (Seca, Hamburg, Germany) at baseline and after the 12-week intervention. BMI was calculated as weight in kg divided by height in meters squared.

Assessment of outcomes

Wound healing and insulin metabolism were considered as the primary outcomes and other metabolic profiles were considered as the secondary outcomes measurements.

Clinical assessment

Ulcer size was determined by multiplying the largest by the second largest perpendicular diameter of the skin lesion. Cumulative ulcer size (sum of the two largest perpendicular diameters and ulcer depth) was also computed for each patient. Depth of the ulcer was defined as either superficial or deep.

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Biochemical assessment

Ten milliliters fasting blood samples at pre- and post-intervention were obtained from each patient. Serum insulin and high sensitivity C-reactive protein (hs-CRP) concentrations were evaluated using ELISA kits (DiaMetra, Milano, Italy) and (LDN, Nordhorn, Germany) with inter- and intra-assay coefficient variances (CVs) below 7%. The homeostasis model of assessment-insulin resistance (HOMA-IR) and the quantitative insulin sensitivity check index (QUICKI) were determined according to the suggested formulas.[25] Hemoglobin A1c ($HbA1_C$) levels in the whole blood were quantified by Glycomat kit (BiocodeHycel, Massy, France) using the method of exchange chromatography. Enzymatic kits (Pars Azmun, Tehran, Iran) were used to quantify magnesium, fasting plasma glucose (FPG) and lipid profiles with intra-assay CVs below 5%. Erythrocyte sedimentation rate was quantified using Westergren method. The plasma nitric oxide (NO) using Griess method,[26] total antioxidant capacity (TAC) by the method of ferric reducing antioxidant power developed by Benzie and Strain,[27] total glutathione (GSH) using the method of Beutler et al.[28] and malondialdehyde (MDA) concentrations by the thiobarbituric acid reactive substances spectrophotometric test [29] were evaluated with inter- and intra-assay CVs below 5%.

Sample size

The sample size was calculated for primary outcome, HOMA-IR: to obtain an effect size of 1.30 and a SD of 1.62 [20], a total sample size of 50 patients (about 25 per arm) was necessary to reach

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a statistical power of 80% considering an overall type 1 error of 5%. Assuming 20% dropouts in each group, the final sample size was determined to be 30 participants in each group.

Randomization and blinding

A computer-generated randomization sequence was developed by an expert statistician using blocks of various length in random sequence. The bottles containing magnesium, vitamin E and placebos capsules were identical and were prepared by another person who did not take part in the investigation and were labeled with patient identification number. Participants and researchers dispensing the capsules and involved in the trial were blinded to the bottle content. All laboratory analyses were blind procedures.

Statistical methods

Anthropometric measures and dietary intakes were compared between intervention groups, using independent samples *t*-test. Multiple linear regression models were used to assess treatment effects on the study outcomes. The effect sizes were presented as the mean differences with 95% confidence intervals. The normality of model residual was tested using the Kolmogorov-Smirnov test. The P-value of <0.05 was considered as statistically significant. All statistical analyses were conducted using the Statistical Package for Social Science version 18 (SPSS Inc., Chicago, Illinois, USA).

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RESULTS

Among participants in the placebo group, 2 participants [withdrawn due to personal reasons (n=2)] and in the supplements group, one participant [withdrawn due to personal reasons (n=1)] were excluded **(Fig.1).** At the end, 57 participants with DUF [magnesium plus vitamin E (n=29) and placebo (n=28)] completed the trial. No side effects were reported after magnesium plus vitamin E co-supplementation in patients with DFU throughout the study.

Gender distribution, mean age, height, weight and BMI at baseline and end-of-trial were not statistically different between the two groups **(Table 1).**

We found no significant difference in mean dietary macro- and micro-nutrient intakes between the two groups (Data not shown).

Compared with the placebo, taking magnesium plus vitamin E supplements reduced ulcer length (β -0.56 cm; 95% CI, -0.92, -0.20; P=0.003), width (β -0.35 cm; 95% CI, -0.64, -0.05; P=0.02) and depth (β -0.18 cm; 95% CI, -0.33, -0.02; P=0.02) **(Table 2).** In addition, co-supplementation led to a significant reduction in FPG (β -13.41 mg/dL; 95% CI, -20.96, -5.86; P=0.001), insulin (β -1.45 μIU/mL; 95% CI, -2.37, -0.52; P=0.003), HOMA-IR (β -0.60; 95% CI, -0.99, -0.20; P=0.003) and HbA1c (β -0.32 %; 95% CI, -0.48, -0.16; P<0.003), and a significant elevation in QUICKI (β 0.007; 95% CI, 0.003, 0.01; P<0.001) compared with the placebo. Additionally,

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compared with the placebo, taking magnesium plus vitamin E supplements decreased triglycerides (β -10.08 mg/dL; 95% CI, -19.70, -0.46; P=0.04), VLDL- (β -2.01 mg/dL; 95% CI, -3.94, -0.09; P=0.04), LDL-cholesterol (β -5.88 mg/dL; 95% CI, -11.42, -0.34; P=0.03), hs-CRP (β -3.42 mg/L; 95% CI, -4.44, -2.41; P<0.001), ESR (β -14.54 mm/h; 95% CI, -19.73, -9.35; P<0.001) and MDA (β -0.30 µmol/L; 95% CI, -0.45, -0.15; P<0.001), and increased HDL-cholesterol (β 2.62 mg/dL; 95% CI, 0.60, 4.63; P=0.01) and TAC levels (β 53.61 mmol/L; 95% CI, 4.65, 102.57; P=0.03). Co-supplementation did not affect other metabolic outcomes.

DISCUSSION

In the current study, we evaluated effects of magnesium and vitamin E co-supplementation on wound healing, glycemic control, lipid profiles, and biomarkers of inflammation and oxidative stress among patients with DFU. We found that magnesium and vitamin E co-supplementation for 12 weeks to patients with DFU had beneficial effects on parameters of ulcer size, glycemic control, triglycerides, VLDL-, LDL- and HDL-cholesterol, hs-CRP, ESR, TAC and MDA levels, but did not affect other metabolic profiles.

Effects on ulcer size parameters

T2DM as a chronic inflammatory state is accompanied by a number of metabolic disturbances. The majority of diabetic patients with DFU present hyperglycemia, insulin resistance and dyslipidemia which lead to the development and progression of further complications.[30] In the

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current study, we observed that magnesium and vitamin E co-supplementation provided a significant improvement in ulcer length, width and depth. There are limited human studies investigating the influences of magnesium or vitamin E supplementation on wound healing. In accordance with the results of present data, our supporting study demonstrated that 12 weeks' consumption of magnesium supplements had beneficial effects on wound healing in DFU patients.[10] In addition, vitamin E supplementation to diabetic rat led to a significant decrease in wound area parameter.[31] In a study by Xu et al.[32], vitamin E administration improved *in vitro* and *in vivo* wound healing. DFU is associated with the elevated risk of amputation and higher mortality rate.[2] The evidence suggests that magnesium acts as an important cofactor for enzymes involved in protein and collagen synthesis during the process of wound curing.[17] In addition, vitamin E intake may improve diabetic wound healing through the regulation of apoptosis, inflammatory response and oxidative stress,[18] enhancing mitochondrial function and increasing the gene expression of cell growth factors.[32] Joint magnesium and vitamin E supplementation may confer more beneficial effects by adding the potential influences related to their single supplementation.

Effects on glycemic controls and lipid profiles

Our findings indicated that magnesium and vitamin E co-supplementation significantly improved FPG, insulin levels, HOMA-IR, QUICKI, HbA1c, and some lipid profiles including triglycerides, VLDL-, LDL- and HDL-cholesterol, but did not affect total cholesterol levels. In

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agreement with the present study, a 4-month magnesium supplementation to prediabetic patients with hypomagnesemia significantly decreased FPG, insulin and triglycerides, and significantly increased HDL-cholesterol levels.[33] Moreover, a 3-month supplementation with vitamin E and/or vitamin C to male individuals with T2DM which were under metformin treatment led to a significant reduction in FPG, insulin levels and HOMA-IR, and a significant rise in QUICKI.[34] Moreover, co-administration of magnesium and vitamin E supplements to obese rat reduced triglycerides levels better than vitamin E alone.[14] Earlier, we reported that a 6-week magnesium and vitamin E co-supplementation to women with gestational diabetes mellitus had beneficial effects on glucose homeostasis markers, triglycerides and VLDL-cholesterol concentrations.[35] However, in contrast to the results of present study, glycemic and lipid parameters were not affected by a 24-week vitamin E supplementation in T2DM patients,[36] and a 8-week magnesium supplementation in healthy young men with a family history of metabolic syndrome.[37] In diabetic patients, hyperglycemia is the most common factor in the onset and progression of DFU, because it impairs with the inflammatory process and induces oxidative damages.[38] In addition, dysregulated both glycemic control and lipid profiles mediate micro- and macrovascular alternations and further co-morbidities.[39] Therefore, the management of hyperglycemia, insulin resistance and dyslipidemia can decrease the burden of these complications. Magnesium regulates insulin secretion form pancreatic beta-cells by the regulation of intracellular calcium homeostasis.[40] It also acts as cofactor for tyrosine kinase enzyme which is vital for the translocation of glucose transporter 4.[40] In addition, magnesium may improve glucose uptake by

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increasing gene expression of peroxisome proliferator-activated receptor gamma.[41] The beneficial effects of magnesium intake on blood lipid parameters may be explained by the regulatory roles on the activity of lipoprotein lipase, fatty acid desaturase and lecithin-cholesterol acyl transferase which are involved in the maintaining lipoprotein balance and fatty acid oxidation.[42] Moreover, vitamin E induces adiponectin expression which involved in the regulation of glucose and lipid metabolism.[43] In addition, vitamin E mediates carbohydrate metabolism through the increasing of intracellular magnesium and glutathione levels.[44]

Effects on biomarkers of inflammation and oxidative stress

The present study showed that taking magnesium and vitamin E co-supplements by DFU patients significantly improved hs-CRP, ESR, TAC and MDA, but did not affect NO and GSH levels. In line with our findings, a 24-week supplementation with magnesium, zinc and chromium significantly reduced CRP levels in subjects with metabolic syndrome.[45] Similarly, the consumption of magnesium supplements decreased hs-CRP and enhanced TAC levels in patients with DFU.[10] In addition, magnesium supplementation in streptozotocin induced diabetic rats lowered MDA levels.[46] Moreover, combined vitamin E and vitamin C supplementation for 12 weeks following acute ischemic stroke led to a significant increase in TAC and a significant decrease in CRP and MDA concentrations.[47] Moreover, the ingesting of vitamin E supplements in patients with non-alcoholic fatty liver disease for 12 weeks reduced CRP and MDA levels.[48] Although in some studies, magnesium or vitamin E supplementation did not change CRP levels

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significantly.[49, 50] The presence of inflammation and oxidative status in diabetic patients is correlated with the development of vascular complications and acceleration of insulin resistance. Previous evidence suggested that the enhancement of antioxidant status and alleviating of inflammatory markers potentially improve endothelial function and blood pressure.[51] The ability of magnesium to diminish oxidative stress and inflammation is due to the activation of antioxidant enzymes, the enhancement of endothelial nitric oxide bioavailability and inhibiting the activity of oxidant enzymes such as myeloproxidase and nicotinamide adenine dinucleotide phosphate oxidase.[52] Vitamin E plays an important role in the scavenging free radicals and by which functions as direct antioxidative agent and protects lipid from peroxidation. It also contributes to the regeneration of other antioxidants. In addition, vitamin E may decrease inflammatory markers by inhibiting nuclear factor kappa B and protein kinas C activity.[53]

The current study had a number of strengths. Firstly, we focused on interesting questions using a randomized, double-blind, placebo-controlled trial. The findings of improved parameters of ulcer size and some metabolic profiles in the magnesium plus vitamin E group in our study are interesting, but need to be confirmed in a larger study. Another strength of the current study was low dropout rate. The current study had few limitations. Firstly, we did not evaluate circulating levels of vitamin E at baseline and end-of-trial. Another limitation was that we could not assess gene expression related to insulin, inflammation and oxidative stress to explore possible mechanisms in patients with DFU. In addition, one cannot conclude if the treatment effects

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observed in the current study was due to the effect of which component of the supplements. Therefore, further studies are needed with single supplements used in the current study in order to evaluate the beneficial effects on parameters of ulcer size and metabolic profiles. Also, further studies are needed to measure parameters of ulcer size by planimetry or digitising techniques compared with one supplement or the other vs. both together.

Overall, magnesium and vitamin E co-supplementation for 12 weeks to patients with DFU had beneficial effects on parameters of ulcer size, glycemic control, triglycerides, VLDL-, LDL- and HDL-cholesterol, hs-CRP, ESR, TAC and MDA levels, but did not affect other metabolic profiles.

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Authors' contributions

ZA contributed in conception, design, statistical analysis and drafting of the manuscript. HA, AJ-K, MM-H, RR, EA, FB and HG contributed in data collection and manuscript drafting. All authors approved the final version for submission. ZA supervised the study.

Conflicts of interest

None declared.

Clinical trial registration number

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Table 1. General characteristics of study participants

	Placebo group (n=28)	Magnesium plus vitamin E group (n=29)	P[1]
Gender (%)			
Male	22 (78.6)	23 (79.3)	1.00†
Female	6 (21.4)	6 (20.7)	
Age (y)	55.5±4.9	57.2±11.0	0.44
Height (cm)	169.1±6.5	171.4±5.7	0.15
Weight at study baseline (kg)	84.9±11.4	88.6±6.6	0.14
Weight at end-of-trial (kg)	85.2±11.4	88.7±6.8	0.18
Weight change (kg)	0.3±1.6	0.1±1.5	0.44
BMI at study baseline (kg/m^2)	29.7±3.9	30.3±3.9	0.57
BMI at end-of-trial (kg/m^2)	29.8±4.0	30.3±3.8	0.66
BMI change (kg/m^2)	0.1±0.6	0.01±0.5	0.35

Data are means± SDs.

[1] Obtained from independent *t*-test.

† Obtained from Pearson Chi-square test.

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Table 2. Wound healing and metabolic profiles at baseline and after 12 weeks of the intervention in patients with diabetic foot ulcer

Variables	Placebo group (n=28)		Magnesium plus vitamin E group (n=29)		Difference in outcome measures between magnesium plus vitamin E and placebo treatment groups[1]	
	Baseline	Week 12	Baseline	Week 12	β (95% CI)	P[2]
Magnesium (mg/dL)	1.51±0.15	1.50±0.15	1.55±0.18	1.83±0.20	0.29 (0.24, 0.34)	<0.001
Ulcer length (cm)	3.1±1.1	2.3±1.3	2.8±1.3	1.6±1.1	-0.56 (-0.92, -0.20)	0.003
Ulcer width (cm)	2.5±0.7	1.8±1.0	2.1±1.3	1.2±0.9	-0.35 (-0.64, -0.05)	0.02
Ulcer depth (cm)	1.1±0.5	0.9±0.5	0.9±0.4	0.4±0.3	-0.18 (-0.33, -0.02)	0.02
FPG (mg/dL)	164.1±39.3	156.0±38.4	173.7±60.8	149.9±47.5	-13.41 (-20.96, -5.86)	0.001
Insulin (μIU/mL)	13.4±3.2	12.7±2.8	14.3±3.8	11.9±3.1	-1.45 (-2.37, -0.52)	0.003
HOMA-IR	5.3±1.6	5.0±1.3	6.0±2.3	5.0±1.9	-0.60 (-0.99, -0.20)	0.003
QUICKI	0.30±0.01	0.30±0.01	0.29±0.01	0.30±0.01	0.007 (0.003, 0.01)	<0.001
HbA1c (%)	7.6±0.6	7.4±0.5	7.4±0.8	6.8±0.6	-0.32 (-0.48, -0.16)	<0.001
Triglycerides (mg/dL)	139.5±38.9	137.1±45.4	156.5±40.6	141.7±36.5	-10.08 (-19.70, -0.46)	0.04
HDL cholesterol (mg/dL)	27.9±8.0	27.4±9.1	31.3±8.1	28.3±7.3	-2.01 (-3.94, -0.09)	0.04
Total cholesterol (mg/dL)	154.8±40.7	151.6±42.1	166.7±50.5	156.8±43.4	-4.88 (-11.47, 1.70)	0.14

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LDL-cholesterol (mg/dL)	91.6±35.7	88.4±37.0	100.8±47.7	90.7±41.3	-5.88 (-11.42, -0.34)	0.03
HDL-cholesterol (mg/dL)	35.4±4.9	35.7±6.1	34.5±9.5	37.7±6.8	2.62 (0.60, 4.63)	0.01
hs-CRP (mg/L)	13.9±4.9	13.6±5.0	10.2±5.0	6.9±4.2	-3.42 (-4.44, -2.41)	<0.001
ESR (mm/h)	30.8±17.2	26.7±17.5	23.8±18.1	8.3±5.8	-14.54 (-19.73, -9.35)	<0.001
NO (µmol/L)	45.3±3.5	45.8±4.5	46.9±1.8	48.3±1.7	1.23 (-0.09, 2.56)	0.06
TAC (mmol/L)	1163.0±136.3	1106.8±153.0	1108.1±68.5	1118.7±73.0	53.61 (4.65, 102.57)	0.03
GSH (µmol/L)	565.7±95.9	550.4±75.7	624.2±116.2	611.7±96.6	25.34 (-8.03, 58.72)	0.13
MDA (µmol/L)	3.2±0.7	3.1±0.6	2.8±0.2	2.5±0.2	-0.30 (-0.45, -0.15)	<0.001

Data are mean ±SDs.

[1]"Outcome measures" refers to the change in values of measures of interest between baseline and week 12. β [difference in the mean outcomes measures between treatment groups (magnesium plus vitamin E group = 1 and placebo group = 0)].

[2] Obtained from multiple regression model (adjusted for baseline values of each biochemical variables, age and baseline BMI).

ESR, erythrocyte sedimentation rate; FPG, fasting plasma glucose; GSH, total glutathione; HOMA-IR, homeostasis model of assessment-estimated insulin resistance; hs-CRP, high-sensitivity C-reactive protein; MDA, malondialdehyde; NO, nitric oxide; QUICKI, quantitative insulin sensitivity check index; TAC, total antioxidant capacity.

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Legend to figure:

Fig. 1. Summary of patient flow diagram

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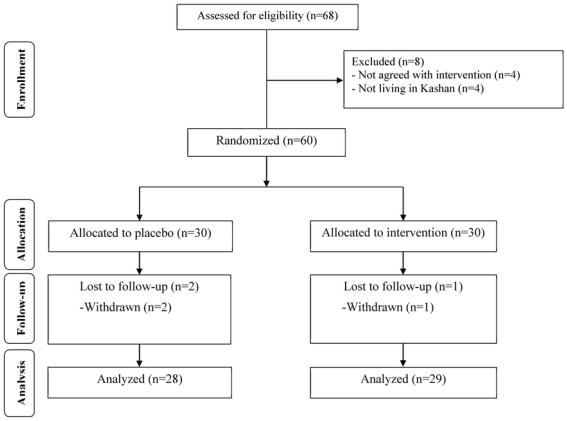

Fig. 1

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